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Remote monitoring system for older adults at risk for complications: a scoping review

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Abstract

Introduction Aging populations and the increasing incidence of chronic diseases are placing additional pressure on healthcare systems. Remote monitoring systems (RMS) have emerged as a solution for improving remote healthcare efficiency and monitoring complex elderly patients. However, telemonitoring services in elderly care are still in their infancy, and their effectiveness remains unproven, with the potential for context-dependent variability. To bridge this gap, we conducted a scoping review to understand the current state of RMSs for elderly individuals at risk for complications, and their potential impact on healthcare service use, elderly's quality of life, and cost-effectiveness.

Methods Following the PRISMA-ScR guidelines, data were gathered from PubMed, Scopus, and Google Scholar databases using predefined keywords, from 2010 onwards. Only English peer-reviewed studies that met the eligibility criteria were retained.

Results A total of 1,576 articles were initially identified. After removing duplicates and screening titles and abstracts, 47 articles were eligible for full-text assessment. Of these, 18 articles were included in the final analysis. Multiple RMSs have been identified and used for elderly patients with complex chronic conditions and a high risk of complications. Such systems typically involve telemonitoring devices, integrated or not with health questionnaires, supported by an automated alert system led by a healthcare professional, usually a nurse, who collaborates with different healthcare structures. The evaluation of RMSs effectiveness was primarily based on health service utilization—particularly hospitalization, emergency room visits and length of hospital stay—followed by quality of life and cost-effectiveness. Our findings support the promising impact of RMSs on reducing hospitalization for at-risk elderly individuals, especially the unplanned hospitalization, improving quality of life, and its cost-effectiveness.

Conclusion This review outlines multiple RMSs used for elderly individuals at risk for complications. Although the effectiveness of RMSs may depend on the content and level of responsiveness, our review underscores the necessity for further empirical research into telemonitoring interventions to fully understand their impact on elderly health outcomes and healthcare systems resources.

Keywords Remote monitoring systems, At-risk elderly individuals, Chronic conditions, Hospitalization, Quality of life, Cost-effectiveness

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Introduction

Healthcare systems in developed countries are increasingly challenged by aging of populations and the growing incidence of chronic diseases. It is estimated that by 2050, the population segment aged 60 and above will double, representing more than one-fifth of the total, and the subset aged 80 and over will reach 20% within this group [1]. In France, according to INSEE [2] 25.6% of the population (i.e., about 17 million people) is over 60, a number expected to rise to nearly 24 million by 2050. This demographic shift poses various social and economic challenges, putting additional stress on healthcare systems that are already stretched to the limit, both in Europe and worldwide. The demand for, and complexity of, healthcare services, including medical care, home care and specialized treatments, is constantly increasing [3]. Elderly individuals generally suffer from multiple chronic illnesses such as heart failure, hypertension, malnutrition, diabetes, and others, often requiring numerous medications [4, 5]. Hence, healthcare expenditure increases with age, particularly in the last two years of life: elderly people are the most common users of both initial and subsequent hospitalizations [6]. Hospital readmission risk also tends to increase, particularly in people aged 75 and over with previous hospitalization(s), long hospital stays, multimorbidity, complex medication regimens, impaired functional capacity and lack of social support [7, 8]. These observations underscore the importance of identifying high-risk elderly patients and providing continuous, comprehensive care to mitigate these risks. As life expectancies increase, it is important to maximize the health of these added years, emphasizing the ability for individuals to remain at home, to help manage escalating healthcare costs [9].

The coexistence of aging and a rising tally of chronic diseases significantly challenges the sustainability of current healthcare models. Each disease is an independent predictor of morbidity, mortality, impaired functional status, and health service use [10, 11]. The combination of multiple chronic diseases heightens the risk of multi-stage decompensation and presents many diagnostic challenges [12]. Such a condition necessitates interdisciplinary and comprehensive follow-up over time, requiring sufficient capacity and competence within the healthcare system. Therefore, it is crucial to develop efficient, cost-effective services for managing complex chronic diseases in the elderly [13]. In this context, the field of telemedicine has emerged as a tool to enhance the efficiency of remote healthcare delivery: enabling resource sharing and coordination across distances. It facilitates real-time health information exchanges between patients and care providers via digital technologies [14]. Telemedicine's main benefit lies in the early detection of health deteriorations through remote monitoring vital parameters, allowing swift interventions during emergencies. It shows promise for preventing severe health issues among elderly patients with multiple conditions. This approach can be used in an emergency, such as a fall or cardiac arrest [15]. In parallel with technological advances, the use of telemedicine has grown considerably over the years [15]. Previous research in various fields has reported its promising effects on health outcomes [16–19] leading to a growing popularity and adoption of these interventions in healthcare [16]. However, many telehealth evaluations lack the rigor required to conclusively determine their effectiveness and cost-efficiency, and it remains challenging to generalize results across various chronic conditions [20]. Aside, several positive patient experiences with telemedicine interventions have been also observed, including greater feelings of empowerment, increased health competence, enhanced awareness of symptoms and treatment, heightened feelings of safety and self-efficacy, and an enhanced sense of independence [21]. Yet, technological and organizational barriers still impede the broader adoption of such telecare solutions [22, 23].

To summarize, it is difficult to draw general conclusions about the effectiveness, efficiency, and feasibility of implementing telemedicine within the healthcare system. While some telemedicine interventions are effective in specific clinical settings, their benefits vary across different contexts [24]. Given the current pressures on healthcare systems, there is substantial interest in the potential of telemedicine to reduce service use among older individuals with chronic conditions and high complication risks. Concurrently, this technology is believed to enhance care quality and reduce costs. Various telemedicine technologies are available, though no consensus exists on the terminology describing them. This paper focuses on remote monitoring systems (RMSs) as emerging solutions in the elderly care pathway, particularly for those at risk for complications. Remote monitoring systems are non-invasive, stand-alone facilities where patient data, such as biometric readings and symptom reports, are transmitted securely to healthcare providers. The latter review the data on digital platforms, which may include built-in automated alerts, and takes action to optimize patient care. Given these observations, we believe that a scoping review helps elucidate the current state of these initiatives at national and international levels and provide an understanding of their potential impact on healthcare service use, elderly quality of life, and cost-effectiveness.

Materials and methods

In this paper, we aim to address the following questions:

1. What remote monitoring systems are available for elderly individuals at high risk?
2. To what extent implanting these initiatives can reduce the burden on the healthcare system and improve the quality of life of the elderly?

Design

To answer the aforementioned questions, we conducted a scoping review to systematically map the literature on current remote monitoring interventions for elderly patients at high risk. The focus of this review was on peer-reviewed remote monitoring interventions in experimental studies, including randomized controlled trials (RCTs), cohorts, and feasibility studies. A scoping review design was selected as we are addressing a broad and emerging topic. This comprehensive approach allows a thorough understanding of current remote monitoring systems used for managing elderly patients'health, offers insights into how these technologies can be optimized to improve its outcomes in real-world settings. Our goal was to explore and provide a descriptive overview of the available evidence, rather than critically appraise individual studies or synthesize evidence from multiple studies [25]. This study adhered to the "Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA)" extension checklist (PRISMA-ScR, 2018) (supplementary material 1).

Search strategy

We searched for articles focusing on remote monitoring interventions for elderly patients at risk for complications and/or hospitalization. In this study, we are mainly interested in peer-reviewed remote monitoring interventions in experimental studies such as randomized controlled trials (RCTs), cohorts and feasibility studies. We used databases like PubMed (MEDLINE), Scopus (Elsevier's database for peer-reviewed literature in science, technology, medicine, and the social sciences), and Google Scholar, starting from the year 2010. This starting point was selected because the contribution of remote monitoring initiatives for elderly patients in healthcare research has grown considerably since then. Two authors (IS and DT) conducted the data collection and selection, while two others (EM, EC) reviewed the work process and output.

Keywords and eligibility criteria

Using predefined eligibility criteria and structured database searches, we identified peer-reviewed articles for experimental studies on remote monitoring of elderly patients, considering the perspective of elderly people at high risk of complication and/or hospitalization, such as multi-morbidities or previous hospitalizations. We used Medical Subject Headings (MeSH) terms and keywords—such as"remote monitoring,""elderly,""chronic disease,""comorbidities"—combined with Boolean operators ("OR"and"AND") to construct our search queries in PubMed, with similar combinations used for other databases (Tables 1 and 2).

Table 1 Eligibility criteria

Eligibility criteria	
Inclusion criteria	● Studies reporting remote intervention for elderly patients at risk for hospitalization and/or with co-morbidities ● Studies involving decompensation preventative monitoring plan with remote in the follow-up phase ● Abstracts must involve at least 3 different keywords (such as, elderly, remote monitoring, complication, hospitalization…) ● English-language peer-reviewed journal articles (experimental studies, RCTs observational and feasibility studies)
Exclusion criteria	● Conference abstracts, abstracts only of published literature, articles in languages other than English (without available translation), and grey (non-peer-reviewed) literature ● Reviews, case studies ● Other languages than English ● Studies reporting only health promotion initiatives

Table 2 Search queries

Database	Search query	Output
PubMed	(("Digital remote monitoring"OR"Monitoring, Ambulatory"[Mesh] OR"Telemedicine" OR"telehealth"OR"telecare") AND ("elderly patient"OR"older adults"OR"Health Services for the Aged") AND ("poly-pathology"OR"comorbidity"[MeSH Terms] OR"chronic disease"[MeSH Terms] OR"chronic disease"[Text Word]))	**161**
Scopus	(("Digital remote monitoring"OR"Monitoring, Ambulatory"[Mesh] OR"Telemedicine"OR"telehealth"OR"telecare") AND ("elderly patient"OR"older adults"OR"Health Services for the Aged") AND ("poly-pathology"OR"co-morbidities" OR"chronic disease"))	**265**
Google scholar	(("Digital health"OR"telehealth"OR"telecare" OR"tele-surveillance") AND ("remote monitoring") AND ("elderly patient"OR"older adults") AND ("decompensation"OR"complication"OR"chronic diseases"OR"co-morbidities"))	**1150**
Total		**1576**

Study selection

The study selection process followed the PRISMA flow diagram (Fig. 1) [26]. Initially, a total of 1576 articles were identified. After removing duplicates and screening titles and abstracts, 47 articles were deemed eligible for full-text assessment. Finally, 18 articles were selected as they met the inclusion criteria and objectives of the study. The selection process and final output were discussed and approved by all authors.

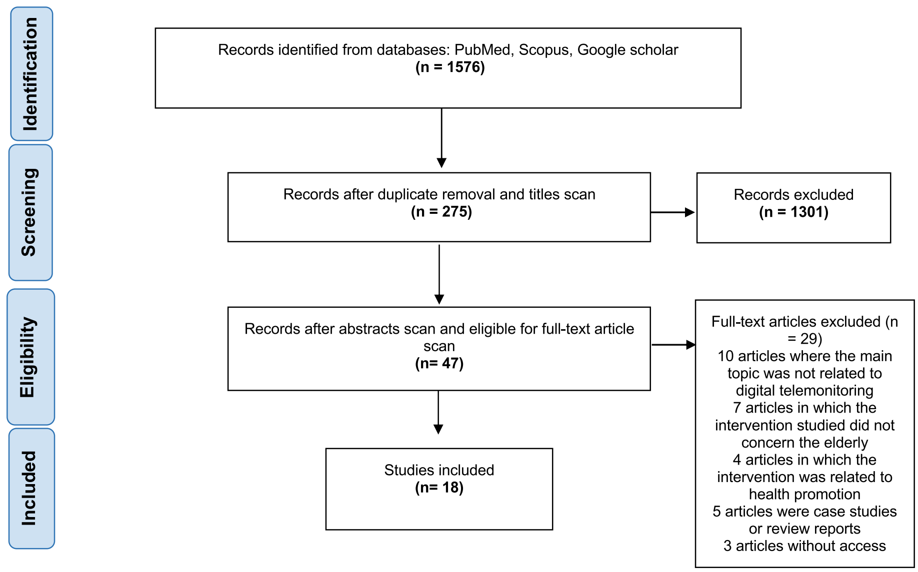

Fig. 1 Study identification, screening, and eligibility based on the Preferred Reporting Items for Systematic. Reviews and Meta-Analyses (PRISMA) guidelines

Data charting process

Data extraction was completed using an Excel file (supplementary material 2). All studies that were deemed to meet the study eligibility after full-text screening were extracted on study characteristics (authors, year, country, DOI, title of study, type of study, target population, date and duration of study, objective of study, selection criteria, setting and location of study, studied intervention, primary endpoint, secondary endpoints, method of sample size calculation, Randomization method (if present), statistical analysis, number of patients included in study, Duration of follow-up, Patient characteristics and group comparability, Primary endpoint results, Secondary endpoint results, Adverse events, Intervention limitations/difficulties) (supplementary material 2). Data were independently reviewed by two authors (IS and DT), followed by multiple discussion meetings for data analysis and synthesis. Given the intent of this scoping review to understand the literature on remote monitoring interventions for the elderly and their potential impacts, no critical appraisal was conducted on the articles included in the study.

Data synthesis

Data were analyzed to identify: firstly, the most common evaluation criteria used to assess the effectiveness of interventions (health outcomes or endpoints); secondly, the potential impact of these interventions in terms of healthcare resources use, quality of life, cost-effectiveness and survival. The results were then summarized in tables (Tables 3, 4, 5, 6, 7 and 8). For this purpose, we pooled results for similar parameters, such as the number of hospitalizations, and described the measurement tools, study results and statistical significance. Organizing the results in this way provided an overview perspective regarding the effectiveness of telemonitoring initiatives in elderly care. The data analysis process was discussed and reviewed by two authors (IS and DT), and the final synthesis was presented to all authors for their input.

Results

Study characteristics

This review included 18 studies, summarized in Tables 3 and 4, conducted between 2012 and 2024. The majority were trials, both pilot and large-scale (14 out of 18), followed by three pre-post longitudinal studies, and one feasibility study. Predominantly, these studies were primarily based in the USA ($n=4$, 22%), and Europe ($n=10$; with Spain and France contributing four (22%) and three (17%) respectively, along with joint studies between Italy and Greece, and recent contribution in Norway and Finland, each contributing one study). Australia and China with two each (11%). The follow-up durations ranged from 3 and 12 months, with most studies ($n=12$, 67%) covering 12 months. Sample sizes ranged from 36 to 636. The mean age of participants, across all the studies, ranged from 55 to 90 years.

Existing Remote monitoring systems (RMSs): state of the art

Multiple telemonitoring systems were identified in the included articles (Table 3). Two randomized controlled trials of interventions were discussed in separate articles. The first, a three-arm RCT discussed in two separate articles, focused on an mHealth app designed to educate users about three common chronic health issues and

Table 3 Summary of identified studies

Authors, year	Country	Study types	Follow-up	Age	Targeted population	Description of identified remote monitoring systems (RMSs)
Gellis et al., 2012 [27]	USA	RCT (2 arms)	12 M	80.1 ± 7.8	Older patients with chronic conditions: HF and/or COPD	Health Monitoring System: tabletop in-home monitor + telemonitoring devices + questionnaire + triage system + telehealth nurse + clinician + integrated electronic medical record + computerized care management tracking tool/**patient conducts self-assessment (measures)/daily**
Takahashi et al., 2012 [28]	USA	RCT (2 arms)	12 M	80.3 ± 8.9	Elderly patients at high score Elder Risk Assessment Index (ERA) > 15	Intel® Health Guid device: real time video conference device + peripheral devices + Nurse led alert system & triage + primary provider (GP, Geriatrics) + assessment + **patient conducts self-assessment (measures)/daily**
De San Miguel et al., 2013 [29]	Australia	RCT (2 arms)	6 M	71 ± 17	Community-Dwelling Older Adults with COPD and at risk for hospitalization	Home-based telehealth equipment (+ questionnaire) + Nurse led alert system + Patient's GP/specialists/**patient conducts self-monitoring/daily**
Pecina et al., 2013 [30]	USA	RCT (2 arms)	12 M	79.8 ± 8.2	Elderly patient at high risk (ERA score)	Intel® Health Guid device: real time video conference device + peripheral devices + Nurse led alert system & triage + primary provider (GP, Geriatrics). Assessment/**patient conducts self-assessment (measures)/daily**
Upatising et al., 2013 [31]	USA	RCT (2 arms)	12 M	80.3 ± 8.9	Elderly patients at high risk for hospitalization (ERA score > 16)	Intel® Health Guid device: real time video conference device + peripheral devices + Nurse led alert system & triage + primary provider (GP, Geriatrics). Assessment/**patient conducts self-assessment (measures)/daily**
Martín-Lesende et al. 2013 [32]	Spain	RCT (2 arms)	12 M	81 ± 7.5	Older Patients diagnosed with HF and/or CLD	Smart phone personal digital assistant (PDA) + Web based platform accessible by professionals (GP, home care nurse..) + telehealth equipment + personalized alerts system/**patient conducts self-monitoring or caregivers/daily**
Doñate-Martínez et al., 2015	Spain	Pre-Post longitudinal study	12 M	65—75	Elderly patients with chronic diseases (co-morbidities) at high risk for hospitalization	Telemonitoring program: equipment for telemonitoring biologic variable and tablet (E-health wireless + alert system + PHCC staff, **patient conducts self-monitoring/daily**
Orozco-Beltran et al., 2017 [33]	Spain	before-and-after study	12 M	70.4 ± 10.4	Patients with chronic health conditions and at high risk of worsening of health condition (CARS scale)	Telemonitoring program: equipment for telemonitoring biologic variable and tablet (E-health wireless + alert system + PHCC nurse, **self-monitoring/daily +** health education and awareness interventions tailored to patient conditions
Mudiyanselage et al., 2018	Australia	RCT (2 arms)	12 M	70.70 ± 11.56	Patient with chronic diseases at high risk for hospitalization	Personalized telehealth monitoring program: remote monitoring platform + telehealth equipment + tablet + individualized care plan (specialist + nurse) + nurse led alert system + response + health coaching sessions (videoconference)/**patient conducts self-assessment (measures)/daily**
Olivari et al. 2018 [34]	Greece and Italy	RCT (2 arms)	12 M	79.6 ± 6.8	Elderly patients with heart failure	Personal health system: wearable wrist + digital devices + alert system + system operators + follow-up center: data were transmitted on a daily basis, 5 days/week, to the eHealth regional center/**patient (or their caregivers) conducts self-assessment (measures)**
Zulfiqar et al., 2020 [35]	France	Feasibility study	3 M	81.4	Polypathologic elderly	Remote telemonitoring: digital platform (MyPredi) + sensors & tablets + questionnaire + professionals (nurse, geriatricians, GP & paramedicals) + alert system/**automated data collection and treatment/daily**
de Batlle et al., 2021 [36]	Spain	RCT (2 arms)	6 M	82 ± 7	Elderly patients with chronic conditions	Personalized care plan: patient self management app (feedback + coach) + monitoring devices, sensor (Fitbit Flex2, oximeter..) + questionnaire SCAM web based E-health platform connecting professionals + case manager + **automated collection/daily**

Table 3 (continued)

Authors, year	Country	Study types	Follow-up	Age	Targeted population	Description of identified remote monitoring systems (RMSs)
Wong et al., 2022, 2023 [37]	China	RCT (3 arms)	3 M	76.56 ± 7.96	Polypathologic Older adults	mHealth app and nurse case management supported by a health-social partnership team + proactive calls (nurse patient: comprehensive assessment, individualized education)/**daily self-assessment**
Gayot et al., 2022 [38]	France	RCT (2 arms)	12 M	87.1 ± 7.58	Polypathologic elderly in nursing home	Telemedicine program: (TELEA) platform + videoconferencing system + equipment for teleconsultation + geriatrician & nursing home staff
Tchalla et al., 2023 [39]	France	RCT (2 arms)	12 M	80.3 ± 8.1	Polypathologic elderly	Remote telemonitoring center: digital platform (e-GEROPASS software: algorithm-based software) + biometric sensors (24/7) + professionals (nurse, geriatricians, GP & paramedicals) coordinating through the software + alert system, **automated collection/daily**
Gahmberg et al. 2024	Norway	RCT (2 arms)	12 M	69.7	Patients with chronic health conditions and at high risk of worsening of health condition,	Remote telemonitoring: telemonitoring devices + tablet to answer questionnaire, monitor data and telecons + follow-up by local primary center + nurse led alert system + individualized care plan by GP after the first few week of monitoring/**patient conducts self-assessment (measures)**
Kokkonen et al., 2024 [40]	Finland	Pre-Post longitudinal study	pre 6 M post 6 M	73 ± 66–80	Older adults with HF at high risk	Veta Health's remote patient monitoring platform (+ questionnaires) + digital devices for measures + automated alert + nurse led alert system & triage/**patient conducts self-monitoring/daily**

RCT Randomized control trial, *HF* Heart failure, *COPD* Chronic obstructive pulmonary disease, *CLD* Chronic lung disease, *GP* General partitioner, *PHCC* Primary healthcare center

Table 4 Primary and secondary outcomes [27–31, 33, 34, 36–42]

			Health services usage					Quality of life (QoL)					Clinical health outcomes			Cost-effectiveness	Patient activation and education	
Authors, Years	Type	Sample size (total) baseline	Nb hospitalization	Nb ED visits	LOS	GP/Specialty	Outpatient visit	EQ-5D-5L	SF-12/SF-36	Other QoL measures	Satisfaction (VAS)	Depression (PHQ)/ (CES-D)/ GDS)	Clinical outcomes	Functional status/ frailty	Death/ survival	ICER/ hosp. cost	Self-efficacy / self-care	Health literacy
Gellis et al., 2012	RCT	*n*= 115	-	X	X	-	X	-	X	-	X	X	-	-	-	-	-	-
Takahashi et al., 2012	RCT	*n*= 205	X	X	X	-	-	-	-	-	-	X	X		X	-	-	-
De San Miguel et al., 2013	RCT	*n*=71	X	X	X	X	-	-	-	X	-	-	-	-	-	X	-	-
Pecina et al., 2013	RCT	*n*= 205	-	-	-	-	-	-	X	-	-	X	-		-	-	X	-
Upatising et al., 2013	RCT	*n*=194	-	-	-	-	-	-	-	-	-	-	X	X	-	-	-	-
Martín-Lesende et al., 2013	RCT	*n*=58	X	X	X	X	X	-	-	-	-	-	-	-	X	-	-	-
Doñate-Martínez et al., 2015	Pre-Post study	*n*= 74	-	-	-	-	-	X	-	-	X	-	-	-	-	-	-	-
Orozco-Beltran et al., 2017	Pre-Post study	*n*=521	X	X	-	-	X	-	-	-	-	-	X	-	-	-	-	-
Mudiyanselage et al., 2018	RCT	*n*= 171	X	-	X	-	-	-	-	X	-	X	X	-	-	X	-	X
Olivari et al. 2018	RCT	*n*= 339	X	X	X	-	-	-	X	-	-	-	-	-	X	-	-	-
De Batlle et al., 2021	RCT	*n*=271	X	X	-	X	X	-	X	-	-	X	X	-	-	X		-
Wong et al., 2022, 2023	RCT	*n*=426	X	X	-	X	X	-	-	X	-	-	-	-	-	-	X	-
Gayot et al., 2022	RCT	*n*=271	X	X	X	X	-	X	-		-	X	-	-	X	X	-	-
Tchalla et al., 2023	RCT	*n*= 534	X	X	-	-	-	-	-	-	-	-	X	-	-	-	-	-
Gahmberg et al., 2024	RCT	*n*= 636	X	-	X	X	X	X	-	-	X	-		-	X	X	X	X
Kokkonen et al., 2024	Pre-Post study	*n*=43	X	X	X	-	X	-	-	-	-	-	-	-	-	X	-	-
Total: 17			12	11	9	6	7	3	4	3	3	6	6	1	5	6	3	2

This table includes 17 out 18 studies as it does not involve feasibility study

Table 5 Hospitalization outcomes

RCTs: Authors, year (sample size: IG vs CG)	Measured outcome	IG	CG	Evaluation	*p*-value	Significant impact
Mudiyanselage et al., 2018 (IG: 86 vs CG: 85)	Hospitalization admission nb (%)	102 (53%)	130 (55%)	Diff: −2%/RRR: 4%	0.813	-
Olivari et al. 2018 (IG: 229 vs CG: 110) [34]	Participant at least one all-cause hospital admission nb (%)	141 (61.6%)	62 (56.4%)	Diff: + 5.2%/RR: 1.1 (0.90–1.33)	0.43	-
	All-cause hospital admissions, mean ± SD	1.48 ± 1.81	1.51 ± 1.91	MD: – 0.02 (– 0.5–0.4)/ RRR: 2%	0.91	-
de Batlle et al., 2021 (IG: 48 vs CG: 28) [36]	All unplanned hospital visits mean ± SD	1.0 (1.1)	2.3 (3.1)	MD: −1.3/RRR: 57%	0.001*	+ (↓)
	Unplanned visits related to chronic disease mean ± SD	0.4 (0.6)	0.9 (1.2)	MD: −0.5/RRR: 56%	0.01 *	+ (↓)
Gayot et al., 2022 (IG: 214 vs CG: 212) [38]	Participant with unplanned hospitalizations nb (%)	50 (23%)	69 (33%)	Diff: −10%, OR = 0.73 (0.43 to 0.97)/RRR: 30%	0.034*	+ (↓)
	Number of unplanned hospitalizations mean ± SD	0.29 (± 0.77)	0.44 (± 0.99)	MD: −0.158/RRR:35%	0.17	-
Tchalla et al., 2023 (IG: 276 vs CG: 276) [39]	Unplanned hospitalizations, nb (%)	108 (40.4%)	130 (48,7%)	Diff: −8.3% age-adjusted and sex-adjusted RR: 0.72, (95% CI 0.51–0.94)	0.043*	+ (↓)
Takahashi Pet al., 2012 (IG: 102 vs CG: 103) [28]	Hospitalization, nb (%)	53 (52.0%)	45 (43.7%)	Diff: 8.2%	0.2359	-
De San Miguel et al., 2023 (IG: 36 vs CG: 35) [29]	Hospitalization nb, mean (SD)	16; 0.44 (0.73)	26; 0.74 (1.2)	−10, −0.03/RRR: 41%	Ns	-
Gahmberg et al., 2024 (IG: 395 vs CG; 241)	Number of unplanned hospital admissions Before and after	0.05	−0.05	DiD: 0.08	0.152	-
	Number of planned hospital admissions Before and after	−0.01	0.01	DiD: 0	0.831	-
Wong et al., 2022. (IGs: 74, 71 vs CG: 76) [41]	Unplanned healthcare utilization total (mean at T2, SE) IG1/IG2 vs CG	0.30 (0.11)	0.47 (0.11)	B: 0.61 (– 1.01 to 2.24)	0.46	-
		0.64 (0.21)	0.47 (0.11)	B: 0.98 (0.32 to 2.09)	0.048*	+ (↓)
Martín-Lesende et al., 2013 (IG:21 vs CG22) [32]	All-cause hospital admissions categorized, 0 admissions, nb (%)/≥ 1 admissions, nb (%)	9 (42.9%)/12 (57.1%)	3 (13.6%)/19 (86.4%)	RR: 0.7 (95% CI 0.4–0.9)	0.033*	+ (↓)
Pre-Post cohorts: Authors, year (sample size)	**Measured outcome**	**Pre**	**post**	**Evaluation: ARR/RRR (95%CI) %**	**p- value**	
Orozco-Beltran et al., 2017 (***n*** = 521) [33]	Hospitalization to an emergency nb (%)	105 (20.2%)	71 (13.6%)	- 6.7 (2–11)/33.2 (11–49)	<.001*	+ (↓)
	Hospitalization for disease exacerbation nb, (%)	55 (10.5%)	42 (8.1%)	- 2.5 (0–8)/23.8 (9–37)	<.001*	+ (↓)
Kokkonen et al., 2024 (***n*** = 43) [40]	Hospitalization HF cause nb, (%)	20 (47%)	6 (14%)	−14/70	.002*	+ (↓)
	Hospitalization all causes nb, (%)	23 (53%)	14 (33%)	−9/39	.08	-

IG Intervention group, *CG* Control group, *RR* Relative risk, *RRR* Relative risk reduction, *Diff* Difference between IG and CG, *DiD* Difference in difference, *MD* Mean difference, *OR* Odd ratio, *ARR* Absolute risk reduction, *NS* Not significant

+: significant impact; (↑: increase or↓:reduction)

self-management techniques, enabling them to record and monitor their conditions through a nurse-led app. This app provides content updates, emergency calls, symptom monitoring, and alerts for abnormal signs, and the nurse acts according to predefined protocols, including counselling, referral to social workers, and escalated care to general practitioners, outpatient clinics, and hospitals [37, 41]. The second intervention, the Tele-ERA study discussed in three articles [28, 30, 31], utilized the Intel® Health Guide device. It involves video monitoring allowing real-time interactions with care providers, supplemented with peripheral devices for vital sign measurements and a questionnaire for daily progress. The majority of included studies targeted elderly individuals with multiple chronic diseases (2 or more) at risk of complications. Five studies specifically targeted older adults with either chronic heart failure or COPD [27, 29, 32, 34, 40]. These were included as we are concerned with the existing interventions of the telemonitoring system that involve elderly patients at risk. Among all the studied included, all but one —the GERONTACCESS study [38], conducted in nursing home settings— focused on home-based remote monitoring. This study was included due to our interest in understanding the French approach to remote monitoring for elderly individuals at high risk. The identified initiatives were predominantly led by nurses and supported by multidisciplinary teams, including hospital physicians, geriatricians, cardiologists, and

Table 6 Emergency department (ED) visits and length of hospital stays (LOS) outcomes

RCTs: Authors, year (sample size: IG vs CG)	Measured outcome	IG	CG	Difference IG vs CG	*P*-value	Significant impact
Mudiyanselage. et al., 2018, (IG: 86 vs CG: 85)	Mean hospital LOS	5.69 ± 16.36	9.58 ± 19.97	−3.89 (−9.40, 1.62)	0.083	-
	Mean hospital LOS (acute admissions)	4.56 ± 11.71	8.66 ± 17.35	−4.1 (−8.56, −0.36)	0.036**	+ (↓)
	Mean hospital LOS (subacute admissions)	1.13 ± 5.20	0.92 ± 4.02	−0.21 (−1.61, 1.19)	0.616	-
Olivari et al. 2018 (IG 229 vs CG 110) [34]	At least one urgent evaluation in ED, Nb (%)	144 (62.9%)	72 (65.5)	RR: 1 (0.81–1.14)	0.73	-
	LOS All-cause hospital admissions (days), mean ± SD	13.13 ± 16.2	16.46 ± 32.05	MD: − 3.33 (− 8.5—1.9)	0.21	-
Gayot C et al., 2022 (IG: 214 vs CG: 212) [38]	LOS mean ± SD	9.37 (± 8.78)	8.14 (± 6.32)	MD: 1.23	0.392	-
	ED visits Nb. (%)	29 (14%)	22 (10%)	4%	0.314	-
Tchalla et al., 2023 (IG: 276 vs CG: 276)[39]	ED visits Nb (%)	16 (6.0%)	6 (2.2%)	+ 4.2%	0.027*	+ (↑)
Takahashi et al., 2012 (IG: 102 vs CG: 103)[28]	ED visit, Nb (%)	36 (35.3%)	29 (28.2) %	7.1%	0.2721	-
	Mean Nb of ED visits per person	0.71 ± 1.3	0.45 ± 0.83	MD: 0.26	0.2293	-
	Mean Nb of days in hospital per person	4.1 ± 8.1	6.1 ± 20.1	MD: −2	0.6055	-
De San Miguel et al., 2023 (IG: 36 vs CG: 35)	ED visits Nb mean (SD)	18; 0.5 (0.77)	21; 0.6 (0.95)	−3, MD: −0.1	-	-
	Hospital LOS (days) Nb, mean (SD)	106; 2.9 (7.3)	183; 5.2 (9.3)	−77, MD: −2.3	-	-
Gellis et al., 2012 (IG: 57 vs CG: 58)[27]	Mean Nb of ED visits	0.6 (1.6)	1.4 (1.2)	−0.8	0.03*	+ (↓)
Martín-Lesende et al., 2013 (IG:21 vs CG22)[32]	Length of stay median (SD)	9.0 (4.3)	10.7 (11.2)	MD: −1.7	0.891	-
	Emergency department attendances median (IQR), year vs Year-1	0 (− 1 to 0)	0 (− 1 to 1)	-	IG:0.210 and CG: 0.981	-
Wong et al., 2022 (IGs: 74, 71 vs CG: 76)[41]	Mean Nb of ED visits at T2 (IG1/IG2 vs CG)	0.03 (0.02)	0.04 (0.02)	β −.01 (− 0.14 to 0.16)	0.89	-
		0.01 (0.01)	0.04 (0.02)	β − 0.05 (− 0.17 to 0.07)	0.42	-
Pre-Post cohorts: Authors, year sample size)	**Measured outcome**	**Pre**	**post**	**RRR (95%CI) %**	**p value**	
Orozco-Beltran et al., 2017 (***n*** = 521) [33]	ED visits Nb (%)	98 (18.8%)	67 (12.8%)	- 32.2 (9–49)	<.001 *	+ (↓)
Kokkonen et al., 2024 (***n*** = 43) [40]	Mean inpatient days all causes, days (95% CI)	2.3 (1.3)	1.2 (1.1)	48	0.17	-
	Mean Nb of ED visit n (95% CI)	1.3 (0.4)	0.7 (0.3)	44	0.006 *	+ (↓)

IG Intervention group, *CG* Control group, *Nb* Number; *RR* Relative risk, *RRR* Relative risk reduction, *Diff* Difference between IG and CG, *DiD* Difference in difference, *MD* Mean difference

+ : significant impact; (↑: increase or↓: reduction)

primary care professionals such as dietitians and social workers. All identified systems employed telemonitoring equipment—primarily peripherals or other connected devices— or the measurement and collection of patient data. In the majority of cases, patients or their caregivers were actively involved in the monitoring process, manually recording vital signs and completing health questionnaires. Only a few studies featured fully automated data collection, either via wearable devices or continuous monitoring systems. The first is the French eCOBALTH study [39], which uses biometric home life analysis technology (biometric sensors 24/7) and tele-homecare automation to monitor and detect early signs of disease decompensation. This system is supported by the e-GEROPASS software (an algorithm-driven decision-support tool) and involves geriatric expertise, with a geriatrician providing guidance to general practitioners and homecare nurses. The second is the Spanish CONNECARE project [36], which combines (1) a web-based Smart Adaptive Case Management (SACM) platform, facilitating streamlined coordination and seamless communication among healthcare professionals; (2) an mHealth self-management system, including a self-management application (with performance feedback, a virtual coach, and the capability for direct communication with the care team), wearables sensors such as the Fitbit Flex 2 digital activity tracker; and (3) a case manager who oversees alerts and serves as the main point of contact for the patient. The third is the French GER-e-TEC study [35], which uses the MyPredi TM e-platform to automatically detect exacerbations of geriatric syndromes, by analyzing data collected through real-time non-invasive connected sensors and health questionnaires (functional and biological), and generating alerts when abnormalities occur. The system also incorporates ambient home sensors facilitating the monitoring of potential risks, such as falls.

Table 7 Quality of life outcome

Authors, year (sample)	Used QoL measures	IG	CG	Evaluation	*P*-value	Significant impact
Gellis et al., 2012 (IG: 57 vs CG: 58)[27]	SF-36 scales: general health (pre-post) M (SD)	Pre: 41.5 (21.1) vs post:48.4 (31.2): 6.9	Pre: 40.6 (21.7) vs post: 41.1 (32.4): 0.5	F value = 3.91	< 0.016*	+ (↑)
	SF-36 scales: social functioning (pre-post) M (SD)	Pre: 46.4 (29.2) vs post: 56.3 (31.9): 9.9	Pre: 45.1 (29.6) vs post: 46.7 (32.3): 1.6	F value = 3.64,	<.0014*	+ (↑)
De San Miguel et al., 2013, (IG: 36 vs CG: 35) [29]	CRQ-SAS only on mastery domain (ability in self- managing their disease)	improved by 2.3	improved by 1.3	to achieve clinical significance a change of 2 is required	*	+ (↑)
Pecina et al., 2013 (IG: 102 vs CG: 103) [30]	SF-12 physical (baseline to 12-month)	−4.3 (9.3)	−1.2 (8.5)	decrease by 12%	0.03*	+ (↓)
Mudiyanselage et al., 2018 (IG: 86 vs CG: 85)	AQoL-8D mean ± SD:	0.63 ± 0.03	0.54 ± 0.03	MD: 0.09 (0.05, 0.14)	< 0.05*	+ (↑)
Olivari et al., 2018 (IG 229 vs CG 110)[34]	SF36v2 PCS (base vs 12 months) mean ± SD:	39.6 ± 9.6 vs 38.2 ± 5.9: −1.4	38.2 ± 9.2 vs 34.1 ± 5.7: 4.1	MD: 2.63 (1.53–3.74)	< 0.0001*	+ (↑)
	SF36v2 MCS (base vs 12 months) mean ± SD:	43.8 ± 10.7 vs 45.1 ± 5.2: 1.3	41.0 ± 10.1 vs 40.6 ± 4.9: 0.4	MD:1.69 (0.07–3.32)	0.04*	+ (↑)
De Batlle J. et al., 2021 (IG: 48 vs CG: 28) [36]	SF-12 physical mean change (baseline vs discharge) (p-value)	MD: + 3.7 (SD 8.4) (p-value: 0.04*)	MD: + 2.0 (7.5) P-value: 0.16	DID: + 1.7 (2.9)	0.21	+ (↑) MD only for IG -
	SF-12 Total score mean change (baseline vs discharge) (P-value)	MD: + 5.8 (SD 12.8) (p-value) 0.03*	MD: + 0.8 (14.7) P-value: 077	DID: + 5.0 (5.2)	0.1	+ (↑) MD only for IG -
Wong et al., 2022 (IGs: 74, 71 vs CG: 76)[41]	SF-12 PCS mean (SE) (T1 vs T2) mean (SE) (T1 vs T2)	IG 1: 39.98 (1.20) vs 42.85 (1.25) IG2: 38.75 (1.08) vs 42.14 (1.04)	39.76 (1.17) vs 41.62 (0.93)	(mHealth: β = 0.22 (− 3.07 to 3.50) mHealth + I: β = − 1.01 [95% CI, − 4.13 to 2.11])	0.90 & 0.53	-
	SF-12 MCS mean (SE) (T1 vs T2) mean (SE) (T1 vs T2)	IG 1: 52.31 (1.29) vs 50.4 (1.20) IG2: 49.71 (1.25) vs 49.51 (1.09)	50.58 (1.32) vs 49.67 (1.25)	(mHealth: β = 1.73 (− 1.89 to 5.34) mHealth + I: β = − 0.87 (− 4.42 to 2.69))	0.35 1 0.63	-
Gayot et al., 2022 (IG: 214 vs CG: 212) [38]	EQ-5D-5L	83 (39)	77 (36)	-	0.60	-
Gahmberg et al. 2024 (IG; 395 vs CG; 241)	EQ-5D-5L before vs after	0.638 vs 0.643	0.659 vs 0.617	DiD 0.05	0.064*	+ (↑)
	EQ VAS: before vs after	54.6 vs 53.9	55.1 vs 48.1	DiD 6.38	0.004*	+ (↑)
Doñate-Martínez et al., 2015 (IG: 74)	EQ-5D-5L Mean (SD)	Pre: 0.63 (0.27)	Post: 0.67 (0.27)	-	0.18	-

IG Intervention group, *CG* Control group, *PCS* Physical Component Summary, *MCS* Mental Component Summary, *RR* Relative risk, *RRR* Relative risk reduction, *Diff* Difference between IG and CG, *DiD* Difference in difference, *MD* Mean difference

+ : significant impact; (↑: improvement; ↓: degradation)

Finally, the study by Olivari et al., 2018 [34], conducted within the "Renewing Health" program, which employed a wrist-worn multi-sensor device ("WristClinic") to o monitor key vital signs including heart rate and oxygen saturation.

Impact on health outcomes and quality of care

The evaluations across the included articles were based on different endpoints, as summarized in Table 4. These evaluation criteria encompassed five major areas: health service usage, quality of life, clinical outcomes, patient activation, and cost-effectiveness. Health service usage was the most frequently assessed outcome ($n = 13$ out of 18 studies), particularly focusing on the number of hospitalizations—including hospitalizations, rehospitalizations, and/or unplanned hospitalizations—used in 12 out of 13 studies, and serving as the primary endpoint in over half of these (n = 7) as represented in Table 4. Emergency department (ED) visits were evaluated in 11 studies, and the length of hospital stays (LOS) was examined in nine studies. Other outcomes were also evaluated, such as outpatient clinics and GP or specialist visits. Quality

Table 8 Cost-effectiveness evaluation

Authors, year, (IG vs CG)	Country	Evaluation	Net benefit	Telemonitoring system cost	cost saving annual and/or per person
De San Miguel et al.,2013 (IG: 36 vs CG: 35) [29]	Australia	DIFFERENCE (TELEHEALTH -CONTROL unit) * unit cost/ per year	224,878 $	$119,350	**105,528 $/per person: 2,931 $ ***
Mudiyanselage et al., 2018 (IG: 86 vs CG: 85)	Australia	ICER based on the incremental cost per QALY between IG and CG	7933 AUD$ per QALY saved	mean total cost between groups + telehealth intervention cost (IG: AUD$12,796 versus control: AUD$12,081)	714 AUD$ (95% CI: 4879, 6308)
de Batlle et al., 2021 (IG: 48 vs CG 28) [36]	Spain	Medical cost CG: 3243.56 vs IG 2482.17	Total medical costs per patient diff: −761.39,	three integrated care (IC) program cost scenarios. (100%: 85.92 US $, 150%: 128.89US $, 200%: 171.84 US $)	**From US $584 to $1434/per patient ***
Gayot et al., 2022 (IG: 214 vs CG: 212) [38]	France	Mean Difference cost (IG-CG) From the perspective of health insurance/MD proportion of patients with unplanned hospitalization	- 350 $/0.091	The telemedicine costs are included in the current care covered by the health insurance	ICER (ΔC/ΔE): $3,846 for each avoided hospitalization
Gahmberg et al., 2024 (IG: 395 vs CG: 241)	Norway	net societal value, per patient for 12 months of follow-up	6530 NOK per patient per year (NOK 7840 – NOK 5220)	42,440 NOK (most likely Most likely scenario)	- 24,640 NOK (−2299 €) per patient receiving follow-up for 12 months (Most likely scenario),
Kokkonen et al., 2024 (pre/post: *n* = 43) [40]	Finland	Absolut change in mean of total cost HSU (pre vs post)	IG €2104 (95% CI €1313-€2895) vs CG €3124 (95% CI €2212-€4036)	-	**−1020€, 33% reduction (*P* = .07) ***

IG Intervention group, *CG* Control group, *ICER* Incremental Cost-Effectiveness Ratio, *QALY* Quality-adjusted life year

Bold*: significant

of life was evaluated in 10 studies, using different measures such as EQ-5D-5L, SF-12 or 36 or other specified measures. Survival or death rates were less commonly investigated, featured in only five studies, despite targeting a population at high risk of complications. The cost-effectiveness of the telemonitoring intervention was also assessed in only 6 studies. Other health outcomes and patient-reported experiences were also evaluated, including the impact on elderly depression, patient efficacy, and satisfaction with RMS. These outcomes were assessed in six, five, and three studies, respectively, using diverse measures, including the PHQ-9 or other Geriatric Depression Scale, the General Self-Efficacy Scale (GSES), the Self-Care of Heart Failure Index (SCHFI), and patient satisfaction surveys.

Synthesis of the main outcomes of included studies

To understand the potential impact of RMSs in high-risk elderly care, we have compiled the results of the most evaluated outcomes in the top three major areas: hospitalizations, ED visits and length of stay for health service usage (Table 5 and 6), the quality of life (Table 7) and cost-effectiveness (Table 8). Each table includes a summary of the overall results and indicates whether significant impact was detected. In addition, supplementary material (2) contains a summary of all study endpoints, including patient satisfaction, self-activation and efficacy, depression, death and/or survival, and other outcomes, to comprehensively elucidate the implications of RMS on elderly health outcomes and overall quality of care.

Health service usage

The impact on health outcomes was particularly evident in a significant reductions in the number of hospitalizations were observed in seven out of the twelve studies that evaluated this outcome, whether rehospitalization, unplanned hospitalization or other type [32, 33, 36, 38–41], De Batlle et al. [36] (sample size IG: 48) and Tchalla et al. [39], (sample size IG: 276) reporting relative risk reductions of 57% and 28% in unplanned hospitalizations, respectively, and a 33.2% reduction in emergency admissions in a cohort study [33] with a sample size of 521 (Table 5). In terms of ED visits and LOS (Table 6), three studies reported significant reductions in ED visits [27, 33, 40] reaching 32% of relative risk reduction in Orozco-beltran et al. [33], contrary to Tchalla et al. [39], showing a significant increase in ED visits. Results on LOS showed no significant reduction, except in the study by Mudiyanselage et al., [43], when hospitalizations related to acute submission particularly. Despite the potential observed improvement in health service usage, the studies reporting survival and death numbers did not indicate any significant improvement [28, 34, 38, 42] (see supplementary material 2).

Quality of Life

Significant improvements in quality of life of RMS groups were noted in six out of 11 studies, on different quality-of-life scales [27, 29, 34, 36, 42, 43]. For instance, Ghamberg et al., a study involving 636 participants over a 12-month follow-up period, detected a significant difference-in-differences (DiD) between the intervention and control groups in both EQ-5D-5L and EQ VAS outcomes [42]. While, De San Miguel et al. [29], showed a significant improvement of 2.3 in the intervention group only on one CRQ-SAS domain, namely the mastery domain (ability to self-manage one's illness). In De Batlle et al. [36], the improvement in quality of life was only significant in the difference between baseline and endpoint results for the intervention group. However, this was not significant in the difference-in-differences (DiD) analysis. Finally, only Pecina et al., [30] reported a statistically significant decrease of 12% in the Physical Component Summary of SF-12 for the intervention group between baseline and endpoint. More detailed study results are summarized in Table 7. These improvements in quality of life were also accompanied by positive patient satisfaction outcomes [27, 42, 44], although it was not significant except in Gahmberg et al. [42], with a DiD of 0.314 (p-value: 0.006). Additionally, there was a significant improvement in depression, as shown by PHQ scores [27, 43] and Geriatric Depression Scale [37] (supplementary material 2).

Cost-effectiveness

The cost-effectiveness evaluations, summarized in Table 8 ($n = 6$), involved annual and/or per person cost-saving evaluation in terms of healthcare usage or incremental cost per Quality Adjusted Life Year (QALY) saved. Three studies reported significant results with savings ranging between approximately €1000 and €2800 per person, including the cost of the telemonitoring system [29, 36, 40]. The other studies also favored RMS as a cost-effective solution, although these results were not statistically significant [38, 42, 43].

Discussion

Remote monitoring systems stand as promising strategies for managing chronic conditions in elderly patients at risk of complications. This scoping review aimed to summarize the current literature on exiting remote monitoring initiatives for older adults with chronic conditions and at risk for complications. We identified 18 empirical studies across various settings and countries, showing encouraging results for telemonitoring in terms of optimizing the use of healthcare services, improving quality of life and enhancing cost-effectiveness.

Remote monitoring systems (RMSs): a minimum of common elements

Our review showed that the identified RMSs differed in terms of their content (i.e., health indicators, monitoring plan, and care management). However, they predominantly incorporated telemonitoring devices integrated with health questionnaires and supported by an automated alert system. A comparative summary is presented in Table 9, which outlines the architecture and key features of each system. The table includes information regarding the use of wearables, the sensors and peripherals employed, the triage method (automated versus manual), the communication technology, and the major functions of the systems.

Our comparative analysis reveals a distinct evolution in the application of telehealth for the management of chronic diseases. Evidence indicated a shift toward hybrid models combining technology-driven data collection and preliminary analysis with clinical oversight by healthcare professionals, typically nurses, who treat data and manage response following predefined protocols, including direct interaction with patient [27–31, 33–37, 39–44]. Thus, the presence of a nurse operator is essential for remote healthcare services, such as videoconferencing and remote monitoring devices, as it ensures regular patient follow-up and early detection of potential health problems, and enables timely, patient-centered interventions. Such a collaboration between machine (i.e., RMS) and professionals (i.e., nurse) who interpret data is pivotal in ensuring the delivery of personalized patient care and follow-up [45]. This involves nurse's key role in improving accessibility and convenience, coordinating with other professionals, and promoting preventive care strategies [46, 47]. Ultimately, nurse-led RMSs may help reduce healthcare costs by minimizing unnecessary hospital visits and readmissions. Notably, patients and their caregivers were actively involved in the telemonitoring process of the majority of the identified RMSs. Their role generally included performing self-assessments and, in some cases, recording and transferring their measurements. Such engagement can enhance confidence in self-management [48] improves individual's safety feeling [49], and raise personal health awareness [30]. However, some challenges were reported, including difficulties managing equipment and mixed perceptions of RMSs usefulness. Patients, especially those with long-standing chronic conditions, may already recognize early signs of decompensation and feel less need for technological support. This may reduce their willingness to engage and accept such novel technologies [29]. Similarly, Wong et al. [37, 41], reported challenges related to the long-term sustainability of mobile health (m-health) applications for chronic conditions self-management. These issues often manifested as app fatigue and declining motivation due

Table 9 Comparatives summary of key features and components of identified RMSs

Study (RMS)	Wearables	Peripherals/Sensors Used	Triage (AI vs Manual)	Communication Tech	Key Features & Functionality	Patient Self-Reporting
Gellis et al., 2012 ("Tele-HEART" via Honeywell HomMed) [27]	**No**	Tabletop telehealth hub with weight scale, BP cuff, pulse oximeter, and thermometer	Nurse-reviewed daily vitals & alerts; **no AI** (no automated risk stratification)	Landline to monitoring center	Daily monitoring with device prompts and educational videos; nurse-led telehealth closely coordinated with home care visits (to manage HF/COPD patients at home)	**Yes** – patients answered daily symptom questions via the telehealth device along with recording vitals
Martin-Lesende et al., 2013 (Basque telemonitoring) [32]	**No**	BP monitor, weight scale, pulse oximeter, peak flow meter	Primary-care nurse triaged alerts; **no** automated stratification	PDA/mobile to web dashboard	Integrated into primary care – daily vitals sent to clinic; family physicians & nurses review data to prevent HF/COPD exacerbations and hospitalizations	**Yes** – patients recorded daily vitals and reported symptoms (e.g. dyspnea) through the system
Olivari et al., 2018 ("Renewing Health" project multicenter EU trial) [34]	**Yes** (wrist-worn "Wrist-Clinic" device measuring HR, SpO_2)	BP monitor, weight scale (± pulse oximeter per site)	Central monitoring center triaged alerts (manual oversight); **no AI** (threshold-based alerts)	Bluetooth to regional center	Daily vital-sign uploads after HF hospitalizations; automatic alerts for threshold breaches; specialized HF nurses monitored dashboard and followed up with patients (education provided via device)	**No** – data collected passively from devices
Kokkonen et al., 2024 (Veta Health's remote patient monitoring platform) [40]	**No** (digital health kit with wireless weight scale and BP monitor)	Digital weight scale (and standard BP cuff for calibration if required) (daily measurements)	Nurse-led data review; **no AI** (standard alerts to clinicians based on thresholds)	Cellular tablet to EMR	Post-hospital HF management with daily weight and BP transmitted to care team; emphasis on timely clinical intervention for any weight/BP changes; HF nurses oversaw a web dashboard and contacted patients as needed	**Yes** – patients actively participated (daily weigh-ins and answering health status questions on the app)
De San Miguel et al., 2013 (COPD telemonitoring) [29]	**No**	Weight scale, BP monitor, pulse oximeter, peak flow meter	Nurse monitored daily data; **no AI** (no automated triage)	Landline auto-upload	Focus on early detection of COPD exacerbation post-hospital discharge; patients took daily measurements and symptom checks (e.g. dyspnea); nurses received alerts for abnormal readings and intervened via phone follow-up	**Yes** – patients reported symptoms daily alongside device-measured vitals
Mudiyanselage et al., 2018 (TELUS Remote Patient Monitoring (RPM) Program)	**No**	Digital health kit: wireless weight scale and BP monitor, HR, glucose, SpO_2	Nurse-led triage with specialist support	Android tablet, dashboard alerts	Real-time data transmission to clinical dashboard; alerts reviewed by nurses; telephone follow-ups; multimedia patient education content provided	**Yes** – daily measurements and symptom reporting
Takahashi et al., 2012 (Intel® Health Guide) [28]	**No**	Multiparameter telehealth hub with weight scale, BP cuff, glucometer, SpO2, and peak flow meter	Nurse monitored data (no automation); **no AI** risk stratification	Cellular tablet + video	Daily vitals tracking via a tabletop telehealth unit; device provided reminders and educational content; nurses reviewed alerts and followed up with patients by phone	**Yes** – patients input daily symptom in addition to automatic vital sign uploads

Table 9 (continued)

Study (RMS)	Wearables	Peripherals/Sensors Used	Triage (AI vs Manual)	Communication Tech	Key Features & Functionality	Patient Self-Reporting
Upatising et al., 2013 (Intel® Health Guide) [31]	**No**	Weight scale, BP cuff, glucometer, SpO2, peak flow meter	Nurse-led review of device alerts; **no** automated stratification	Cellular tablet to clinic	Daily vital signs transmitted via the Intel hub aimed at early detection of decline; periodic nurse check-ins (including video calls) to assess status	**Yes** – patients (often with caregiver help) input daily symptom in addition to automatic vital sign uploads
Pecina et al., 2013 ("Tele-ERA" trial; Intel® Health Guide)) [30]	**No**	BP monitor, weight scale, pulse oximeter (glucometer if diabetic)	Clinician (MD) monitored data; **no AI** (manual oversight of alerts)	Internet + phone/video	Telehealth integrated into primary care for complex patients; daily self-measurements and symptom check-ins were reviewed by clinic staff; aimed to reduce ER visits/hospitalizations. Educational modules provided (with extra support for high-risk patients)	**Yes** – patients self-reported daily wellness (e.g., completed "How do you feel?" surveys) along with measuring vitals
Donato-Martínez et al., 2015 ("Valcrònic" program, Spain)	**No**	Weight scale, BP monitor, glucometer, SpO2, peak flow meter (condition-specific kit)	Primary care nurse case-manages alerts; initial **risk strat** used for enrollment (high-risk patients), but **no AI** in daily triage	Tablet to EMR/dashboard	Large-scale telehealth program in primary care centers; patients submitted daily/weekly readings via tablet which uploaded to an EHR-linked platform; nurses monitored multiple chronic patients and adjusted care or contacted the patient or GP as needed	**Yes** – patients report daily vitals and symptom status
Zulfiqar et al., 2020 ("GER-e-TEC" MyPredi™ platform) [35]	**No** traditional wearables; only a pedometer for activity/sleep tracking;	BP monitor, weight scale, glucometer, SpO2 (devices per patient's conditions) ambient home sensors for safety	**AI-driven triage** – automated system detected risk changes and triggered alerts; **yes** AI risk stratification	WiFi to MyPredi dashboard	AI-powered e-health platform (MyPredi) continuously analyzed 24/7 biometric data; automatic alerts to clinicians when patient risk indicators (e.g. vital trends) worsened. Geriatric care team monitored a dashboard and coordinated interventions with the patient's GP	**No** – mostly passive data collection via sensors
De Batlle et al., 2021 (CONNECARE project) [36]	**Yes** (Fitbit Flex 2 activity tracker)	Bluetooth BP monitor, weight scale, SpO2 sensor linked via patient's smartphone/tablet; (additional devices as needed per patient)	Automated symptom alerts + nurse triage; **yes** (app generates alerts on thresholds, nurse confirms)	Web/app platform	Smartphone app + cloud platform linked to hospital EHR; patients measured vitals and answered app questions daily, data synced to care team's dashboard. Alerts are sent to a nurse case manager. Multidisciplinary team (including specialists) could collaborate via the platform	**Yes** – patients actively used the app to report symptoms and health status each day
Gayot et al., 2022 ("GERONTACCESS" trial) [38]	**No**	Standard vital-sign devices used by nursing staff (BP, glucometer, etc.)	Scheduled teleconsultations (geriatrics <—> nursing home); **no** continuous sensor triage or AI	Video conferencing (TELEA)	Regular virtual visits between nursing home residents and a geriatrician via teleconference; nursing staff facilitated sessions and carried out any clinical recommendations	**No** –passive data collection monitoring was done by staff and doctors

Table 9 (continued)

Study (RMS)	Wearables	Peripherals/Sensors Used	Triage (AI vs Manual)	Communication Tech	Key Features & Functionality	Patient Self-Reporting
Wong et al., 2022; 2023 (Telehealth coaching program, Hong Kong) [37, 41]	**No**	No connected sensors provided (patients self-measured BP, etc., at home as advised)	No automated triage; nurse and peer team manually flagged issues; **partial** risk-based escalation (higher-risk got more attention)	Smartphone + WhatsApp	Nurse-led remote coaching program. Communication via phone calls and messaging (including WhatsApp) provided ongoing support. Peer volunteers were involved to encourage and sustain participant engagement	**Yes** – participants actively reported home measurements and lifestyle activities to the coaches and received feedback
Tchalla et al., 2023 (eCOBAHLT trial) [39]	**No wearables** (continuous biometric monitoring devices for 24/7 vitals)	Home monitoring kit BP, weight, SpO_2, etc. via a tablet gateway; +. ambient home sensors (motion, fall detection)	Clinician-led multi-tier triage + plus **rule-based automated alerts** ("traffic-light" alerts when patient-specific thresholds crossed)	e-GEROPASS + SMS/email	Comprehensive remote monitoring supervised by a geriatric team. Daily vital signs were automatically transmitted; a "traffic-light" system categorized readings (green/yellow/red) based on personalized limits. Geriatric specialists provided feedback and care recommendations to the patient's primary care doctor	**No** – patients/caregivers took required measurements, but patients did not fill out daily symptom diaries (monitoring was driven by sensors and clinician assessment)
Gahmberg et al., 2024 (Norway RCT)	**No**	Connected health devices for vitals (BP, blood glucose, SpO_2, weight) + tele-exam tools (digital stethoscope, camera)	Nurse-led triage with specialist backup; **no AI** (threshold-based alerts reviewed by clinicians)	Tablet + auto-transmit	Primary care nurses and GPs remotely monitored patients' daily vitals through a secure platform.Managing a range of chronic illnesses in the community by intervening early when readings were out-of-range. Integrated within usual primary care workflow	**Yes** – patients report daily vitals and symptom status
Orozco-Beltrán et al., 2017 (ValCrònic program) [33]	**No**	BP monitor, weight scale, glucometer, pulse oximeter (kit tailored to each patient's conditions)	Nurse-led triage based on risk-stratified alerts	Bluetooth tablet to platform	Regular teleconsultations; daily vital signs monitoring via Bluetooth-connected tablet; symptom tracking; automated alerts; tailored educational content	**Yes** – daily symptom questionnaires and vital signs logging

to overfamiliarity with the app features. Addressing these usability and user experience barriers is essential to facilitate the uptake of technology-based health monitoring systems [50, 51].

While, only a few of the identified RMSs have incorporated wearable devices or continuous automated data collection [34–36, 39], it is clear that continuous patient monitoring with the ability to trigger automated alerts via an integrated digital platform, can significantly enhance personalized follow-up and enable timely care response. Wearables and connected devices can report a wide range of physiological variables, including but not limited to heart rate, blood pressure, body temperature, oximetry, weight, electrocardiogram (ECG), and capillary blood sugar. More specific data can also be collected, such as physical activity and sleep patterns using pedometers, as demonstrated in the GER-e-TEC study [35, 52]. Continuous real-time RMSs can be beneficial for older individuals with complex chronic conditions. However, several practical challenges remain, especially related to user compliance and batter life concerns [53]. Elderly patients may encounter physical or cognitive challenges that hinder their ability to use devices effectively. They may face difficulties in maintaining consistent wear and recharging or replacing batteries for sensors or wearable monitors, which can impede long-term utilization [54]. Furthermore, previous studies have identified technical and ethical challenges associated with the extended use of wearables for health purposes, including accuracy, privacy, security, and cyber risks [55–57]. These issues may explain why the majority of reviewed projects chose simpler home monitoring kits over wearables.

Emerging evidence on evaluation Outcomes

This review identified multiple measures used to assess the impact of RMSs on the health outcomes and quality of care of at-risk elderly individuals. Hospitalizations and emergency department visits were the most commonly evaluated health outcomes, followed by hospital length of stay, quality of life, and cost-effectiveness. Our findings align with those of previous reviews on tele-transition interventions for older adults at risk [58]. Other key health outcomes were also identified, such as mortality or survival rates [28, 34, 38, 42], and depression [27, 28, 30, 37, 41, 43], which is commonly associated with chronic diseases and related to excess morbidity and mortality [59]. In addition, specific health-related outcomes were utilized to assess the impact of RMS, including self-efficacy [30, 37, 41, 42], patient activation and health literacy [42, 43]. These are essential for patient self-management, which is a key element in successful remote monitoring interventions [24].

This study does not seek to evaluate or confirm the efficacy of a remote monitoring intervention in elderly care. Rather, our aim was to understand their potential impact on the most studied health outcomes (i.e., hospitalization, ED visits, LOS, quality of life and cost-effectiveness). Overall, the findings were inconsistent across different contexts, yet promising, with the greatest benefit observed in the prevention of unplanned hospitalizations [33, 36, 38, 39, 41]. A continuous monitoring of biometric parameters with a comprehensive evaluation conducted at the older adult's home enables remote care operators (e.g., nurses, physicians, or geriatricians) to manage potential clinical issues before a decompensation and avoid unplanned hospitalization. Meanwhile, ED visits and LOS outcomes were hardly encouraging, only a limited number of studies indicated a notable decline in ED visits [27, 33, 40] and LOS [43]. Contextual and methodological challenges may explain this lack of significance [28]. In the same line, Tchalla et al. [39] attributed the significant increase in ED visits to the unavailability of the treating physician, resulting in an uptick in emergency room admissions for treatment adjustment. Also, the findings of this review support a promising impact of RMS intervention on the quality of life for the elderly at risk. Significant improvements in the intervention group or differences from the control group were reported in terms of quality of life [27, 29, 34, 36, 42, 43]. Sufficient information on vital health and regular check-ups by healthcare professionals would enable the elderly to better manage their chronic illnesses and improve their quality of life [60]. However, in Pecina et al. [30], a significant reduction and deterioration in self-perceived physical health (measured by PCS scores) and no significant improvement in self-perceived mental well-being (measured by MCS scores). These findings were explained by the types of telemonitoring, with the authors suggesting that different populations may benefit more than others.

Finally, this review suggests that RMS may be a cost-effective approach to healthcare, although the evidence remains limited. Some studies have demonstrated cost saved per person on healthcare usage, while others have shown promising numbers for annual cost savings and ICER per QALYs saved, even when including the remote monitoring costs. These savings align with the significant reduction in hospitalization numbers observed in De Battle et al., and Kokkonen et al. [36, 40], as well as the 41% reduction in hospitalization mean observed in De San Miguel et al., although it was insignificant [29]. This is consistent with the findings of another systematic review, which showed that telecare interventions are cost-effective and suitable for large-scale implementation in healthcare facilities [58]. Elderly people, especially those with multiple co-morbidities and functional incapacity, are most vulnerable to deteriorating health status leading to unplanned hospitalization, emergency department visits as well as an augmented hospital stays. Anticipating risks and rapidly managing deteriorating health status thanks to a comprehensive personalized, real-time telemonitoring system seems essential to avoid complex interventions and reduce total healthcare costs.

Design recommendations for RMSs

Our findings suggest that successful remote monitoring interventions are characterized by the implementation of key design principles that enhance usability, support clinical integration, and improve patient outcomes. In light of these findings, we propose a set of recommendations to be considered when designing RMSs for older adults' care:

- Integration with clinical workflow and interoperability: Integrating RMSs into existing care processes, while ensuring interoperability with clinical IT systems, is crucial to ensure providing coordinated and effective care. Seamless integration allows healthcare professionals, usually nurse case manager, to access remote monitoring data alongside other clinical information (e.g., EHR), interpreting alerts and to respond promptly within the usual care continuum.
- Proactive monitoring and early intervention: RMSs must enhance and prioritize early detection of health changes through daily tracking of vital signs (via connected devices or wearables), symptom reporting through questionnaires, and triggering automated alerts drawing on a predefined thresholds protocol or algorithm-based triage. These tools enable prompt intervention that can help prevent deterioration
- Active patient engagement and self-reporting: Both are essential components in a patient-centered care model. Combining automated and/or self-

measurements with self-reported symptoms and educational assistance, can enhance patient engagement and adherence, thereby encouraging self-management

- Usability and simplicity of technology: Designing RMSs with user-friendly interfaces and equipment, combined with automated data uploads tailored to meet older adults needs, can potentially alleviate technology-related burdens and improve adoption.
- Personalization and support networks: These aspects should be carefully considered when designing RMSs by implementing features such as customized monitoring protocols and thresholds, actively involving caregivers or peer supporters, and providing real-time adaptive responses. This approach helps enhance relevance, reduce alert fatigue, and improve patient adherence.

Limitations

This study has some of limitations. First, our search strategy was limited to English-language articles, so it may be possible that relevant studies in other languages were missed. However, it is important to note that almost half of identified articles were based on studies conducted in non-English-speaking countries such as Spain, France and China. In addition, we are interested in peer-reviewed interventions that have demonstrated their contribution at an international level. Second, our keywords selection was limited, and due to the ambiguity of certain concepts related to the research area, in particular the concept of "remote monitoring" it is possible that our search protocol missed pertinent studies. To mitigate potential exclusions, we employed synonyms and Medical Subject Headings (MeSH) terms (e.g., "telemedicine" OR "telehealth" OR"Monitoring, Ambulatory"[Mesh]) to broaden our search scope. Additionally, we examined the reference lists of the articles we included, aiming to capture any significant studies that our database searches might have missed. Third, as the research scope was limited to RMSs targeting patients at risk for hospitalization, with very limited attention to systems within a wellness or primary prevention context, such as robotics in smart home environments, this may explain why none of the identified studies explicitly involved robotic caregivers and only a few projects incorporated *smart-home* elements via ambient sensors usage [35, 39]. Robot-assisted smart homes represent an important emerging frontier in remote care for older adults. This area of research can be explored in future work. Finally, the starting point for our review in 2010 was based on the assumption that the clinical translation of telemonitoring initiatives for at-risk older people was fairly recent. Although this decision was pragmatic, it may have excluded relevant studies. Despite these limitations, this scoping review is pioneering in its examination of remote monitoring systems for elderly individuals at risk for complications. It contributes to the existing literature on telemedicine in elderly care by offering valuable insights into existing RMSs and the diverse impacts of telemonitoring interventions on the elderly. These insights have significant implications for healthcare service utilization, patient quality of life, and cost management.

Conclusion

This review outlined various remote monitoring systems employed to support elderly individuals at risk of complications, analyzing their potential effects on reducing healthcare utilization—specifically focusing on hospitalizations, emergency department visits, and length of hospital stays—alongside their impact on quality of life and economic outcomes. The findings regarding the effectiveness of remote monitoring systems in elderly care show promising yet inconclusive impacts. Our review showed that the content of these RMSs differs between contexts. For a specific population, such as older adults with complex chronic illnesses, the effectiveness of these RMSs depends not only on the monitoring of health indicators, but also on the capacity to provide personalized monitoring and care plans. This requires high-level responsiveness and coordination between multidisciplinary teams at different levels. In light of our findings, we have provided a set of recommendations for designing RMSs for the care of older adults. Overall, this review underscores the necessity for further research into telemonitoring interventions to fully understand their influence on health outcomes for elderly populations with multiple chronic conditions.

Abbreviation

RMS	Remote monitoring system
GP	General practitioner
ED	Emergency department
LOS	Length of stay
IG	Intervention group
CG	Control group
RCT	Randomized control trials
QALY	Quality Adjusted Life Year
ICER	Incremental cost-effectiveness ratio

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12877-025-06385-8.

Additional file 1: Preferred Reporting Items for Systematic Reviews and Meta-Analysis Extension for Scoping review (PRISMA-ScR)

Additional file 2: Study summary

Authors' contributions

EM, and IS conceived the idea. IS, with the help of DT designed the study. IS and DT performed data collection and selection according to inclusion and exclusion criteria. All steps were reviewed and verified by EC and EM. Selected studies were approved by all authors. IS wrote the first draft and collaborated with DT and EM to generate the final draft. EM verified the analytical methods, proposed corrections, and supervised the overall research. All authors provided critical feedback and helped shape the research, analyses, and manuscript. All approved the final version.

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Data availability
All data generated or analyzed during this study are included in this published article and its supplementary information files.

Declarations

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Not applicable.

Consent for publication
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The authors declare no competing interests.

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